

LD10 Master Theory A-to-Z

Boosting theory explained in simple language with business context

1. What is boosting?

Boosting is a machine learning method where models are trained one after another. Each new model focuses on errors made by the previous models. The final prediction is made by combining many small models into one stronger model.

Simple memory

Bagging builds many models independently. Boosting builds models sequentially, where the next model fixes earlier mistakes.

2. Bagging vs Boosting

Topic	Bagging / Random Forest	Boosting
Training	Trees train independently	Trees train one after another
Main goal	Reduce variance	Reduce errors step by step
Example	Random Forest	XGBoost, CatBoost, LightGBM
Risk	Can miss sharp patterns	Can overfit if training is not controlled

3. Weak learners and decision stumps

A weak learner is a simple model that is only slightly better than guessing. In boosting, small decision trees are commonly used. A tree with only one split is called a decision stump. Many weak learners together can create a strong model.

4. AdaBoost

AdaBoost increases the weight of wrongly predicted rows. The next model pays more attention to those difficult rows. In the notebook mini-demo, AdaBoost reached accuracy 0.833 and F1-score 0.845.

5. Gradient Boosting

Gradient Boosting does not only reweight rows. It learns how the current model should be corrected. For regression, the next tree often learns the remaining error between actual values and predicted values.

6. Forward stagewise additive modelling

Forward stagewise additive modelling means building the final model step by step. Each new tree is added to the existing model. The model becomes: first prediction + correction 1 + correction 2 + correction 3.

7. Loss functions and pseudo-residuals

A loss function measures how wrong a model is. A pseudo-residual tells the next model the direction in which the prediction should move to reduce the loss. In simple language, it is the correction signal for the next tree.

8. Learning rate, n_estimators and max_depth

Parameter	Simple meaning	Business risk if wrong
learning_rate	How strongly each tree corrects the model	Too high can overfit; too low can be slow.
n_estimators	Number of boosting trees	Too few underfit; too many can overfit.
max_depth	How complex each tree can be	Deep trees may memorise noise.

9. Early stopping

Early stopping stops training when validation performance no longer improves. It is a practical way to prevent unnecessary model complexity. In the CatBoost section, the best iteration was 442.

10. XGBoost

XGBoost is a strong, regularised version of Gradient Boosting. It is reliable for structured business data and provides useful feature importance. In the Bank Marketing task, it achieved F1-score 0.8499.

11. CatBoost

CatBoost is designed for categorical features. It can use category columns directly when we tell the model which columns are categorical. This reduces manual encoding work.

12. LightGBM

LightGBM is a fast Gradient Boosting library. It is often useful on larger tabular datasets. Important controls include num_leaves, learning_rate and min_child_samples.

13. GridSearchCV vs Optuna

Method	How it searches	Best use
GridSearchCV	Tries fixed combinations from a grid	Small and clear search spaces
Optuna	Uses smarter trial-based search	Larger or more flexible tuning spaces

14. Averaging ensemble

An averaging ensemble combines predictions from several strong models. The idea is that different models make different mistakes. Averaging can improve stability, but it also increases complexity.